Chapter 4 - Applications of Differentiation

Calculus I

November 4, 2024

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Section 1

Local Extrema

Global Extrema

Recall: The Extreme Value Theorem tells us that if $f(x)$ is continuous on $[a, b]$, then there exists $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d)$$

for all $x \in [a, b]$, and each of c and d is either

1. an endpoint, or
2. is in the open interval (a, b).

Local Extrema

Definition 4.1.1

1. We say that $f(x)$ has a local maximum at $x = c$ if there exists an open interval (a, b) containing c such that

$$f(x) \leq f(c)$$

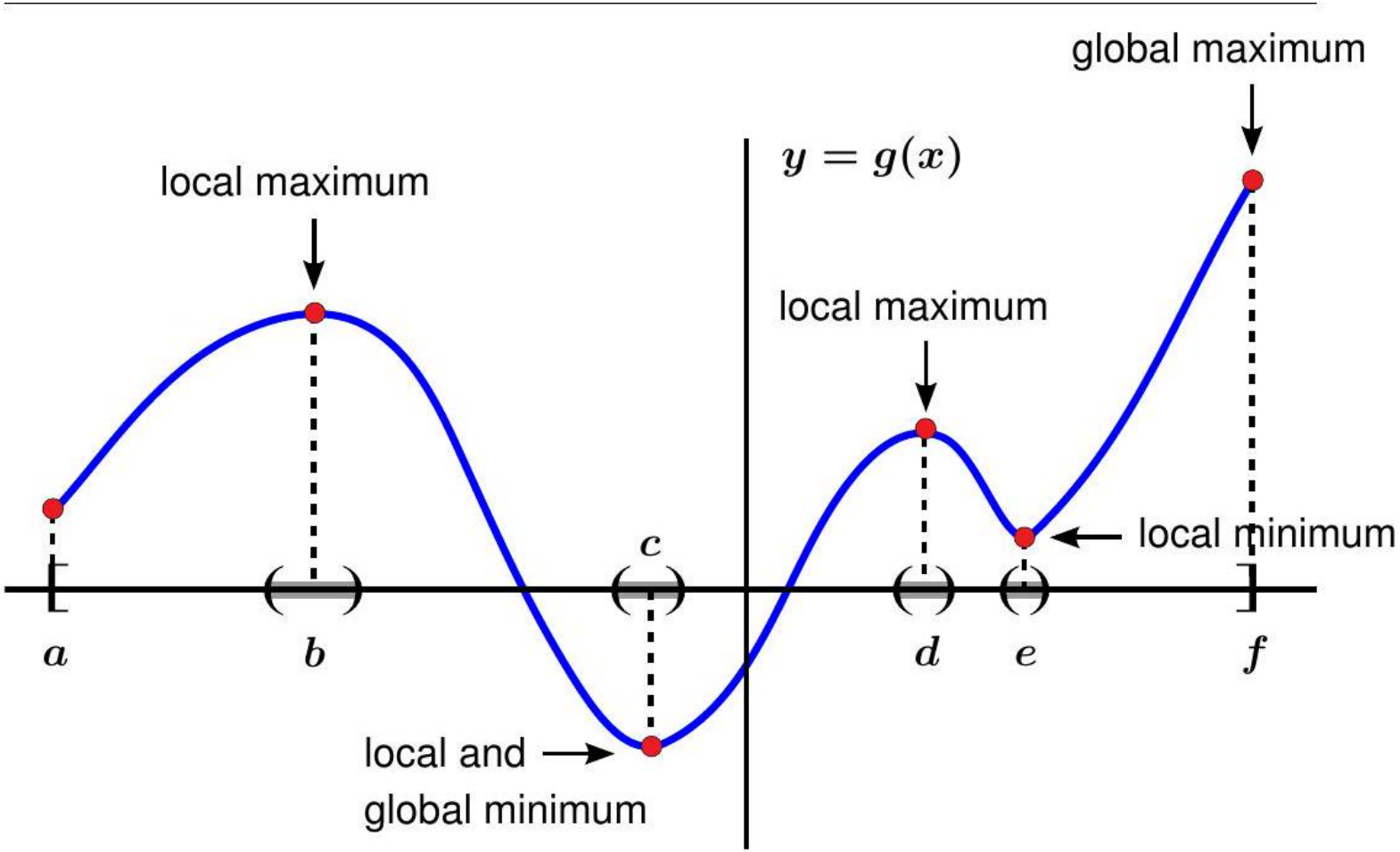
for every $x \in (a, b)$.

2. We say that $f(x)$ has a local minimum at $x = c$ if there exists an open interval (a, b) containing c such that

$$f(c) \leq f(x)$$

for every $x \in (a, b)$.

Local and Global Extrema

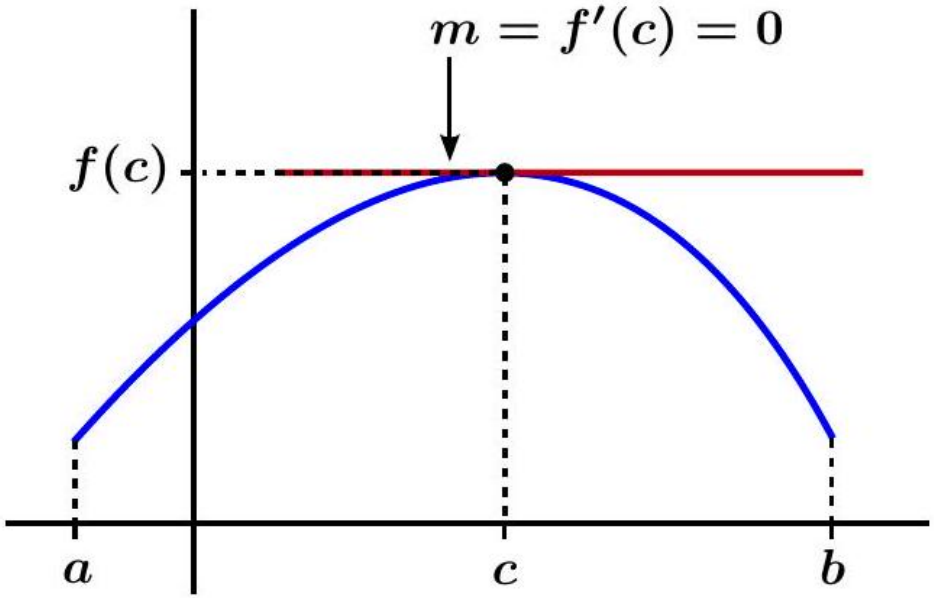


$g(x)$ is defined on $[a, f]$.

Local Extrema

Problem: If f has either a local maximum or local minimum at $x = c$ and f is differentiable at $x = c$, what can we say about $f'(c)$?

Local Extrema

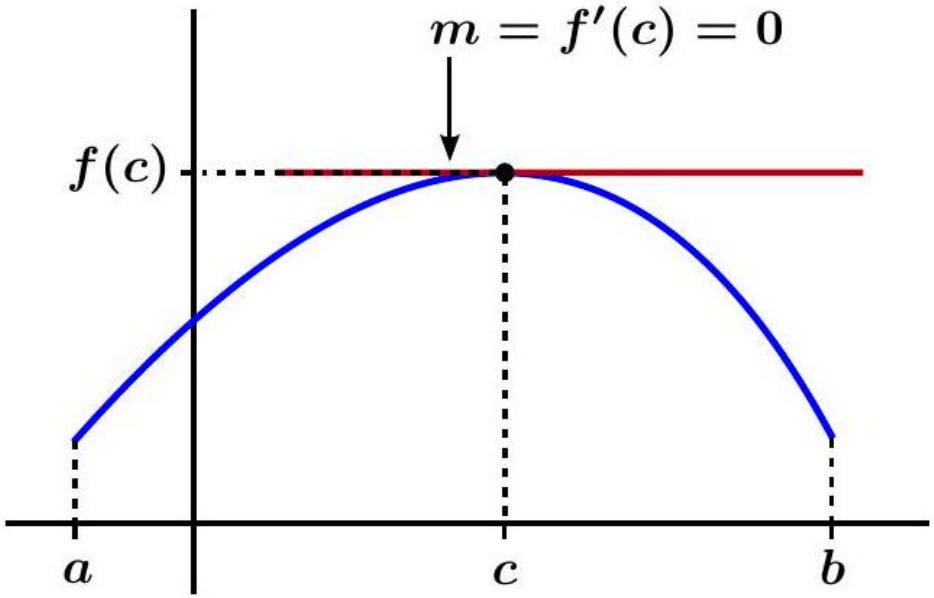


Solution: Assume that $f(x)$ has a local maximum at $x = c$ and that $c \in (a, b)$ with $f(x) \leq f(c)$ for all $x \in (a, b)$. Assume that $f'(c)$ exists.

1. If $a < c + h < c$, then

$$\frac{f(c+h) - f(c)}{h} \geq 0 \Rightarrow f'(c) = \lim_{h \to 0^-} \frac{f(c+h) - f(c)}{h} \geq 0$$

Local Extrema



2. If $c < c + h < b$, then

$$\frac{f(c+h) - f(c)}{h} \leq 0 \Rightarrow f'(c) = \lim_{h \to 0^+} \frac{f(c+h) - f(c)}{h} \leq 0$$

Hence $f'(c) = \mathbf{0}$.

Local Extrema

Theorem 4.1.2 (Local Extrema Theorem)

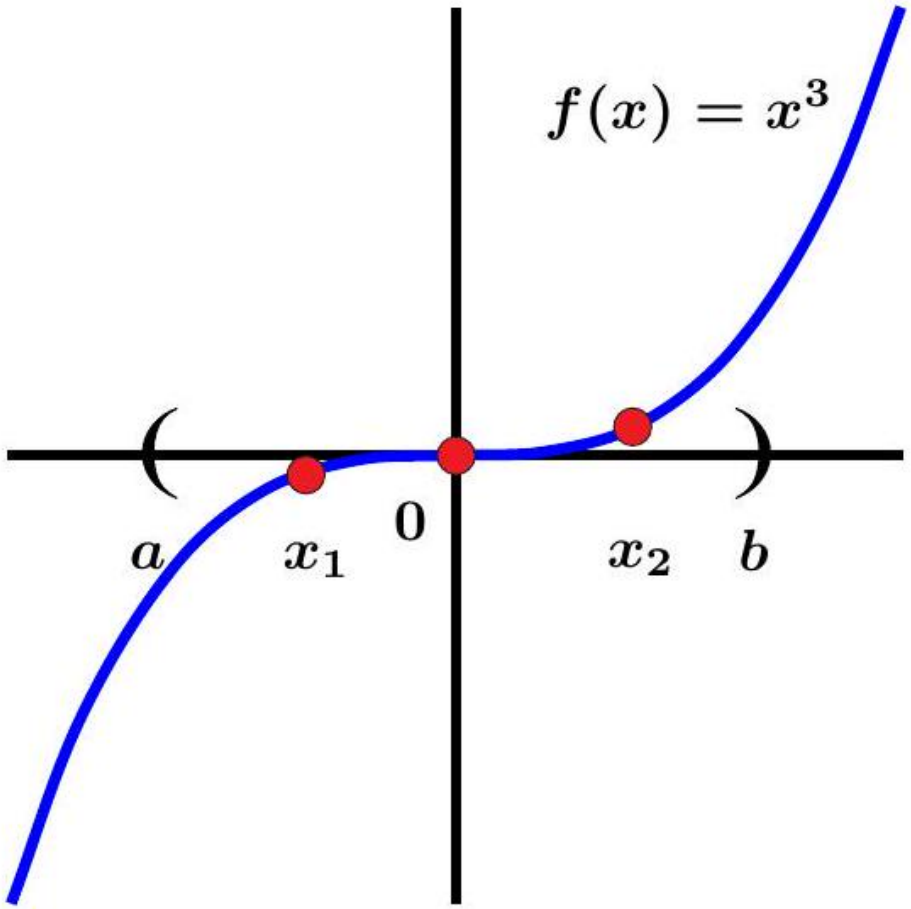
Assume that $f(x)$ has either a local maximum or a local minimum at $x = c$. If $f(x)$ is differentiable at $x = c$, then

$$f'(c) = 0$$

Question:

If $f'(c) = 0$ must $x = c$ be either a local maximum or local minimum?

Local Extrema

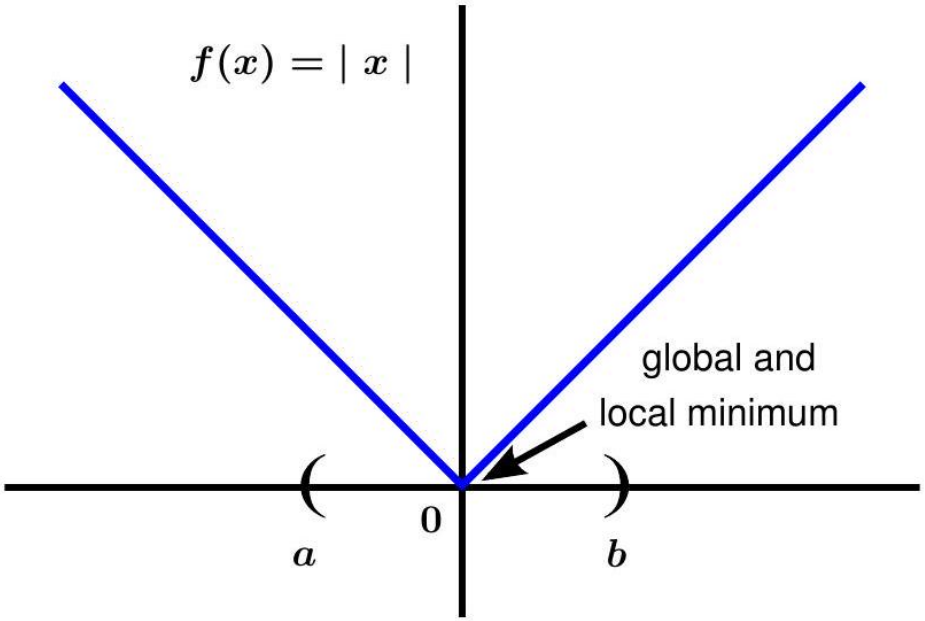


Example: Let $f(x) = x^3$. Then $f'(x) = 3x^2$ so that

$$f'(0) = 0$$

But $f(x)$ has neither a local maximum nor a local minimum at $x = c$ since $f(x)$ is always increasing.

Local Extrema



Question: If $f(x)$ has a local maximum or minimum at $x = c$, must $f'(c) = 0$?

Example: Let $f(x) = |x|$. Then $f(x)$ has a local (and global) minimum at $x = 0$, but $f'(0)$ does not exist.

Critical Point

Definition 4.1.3 (Critical Point)

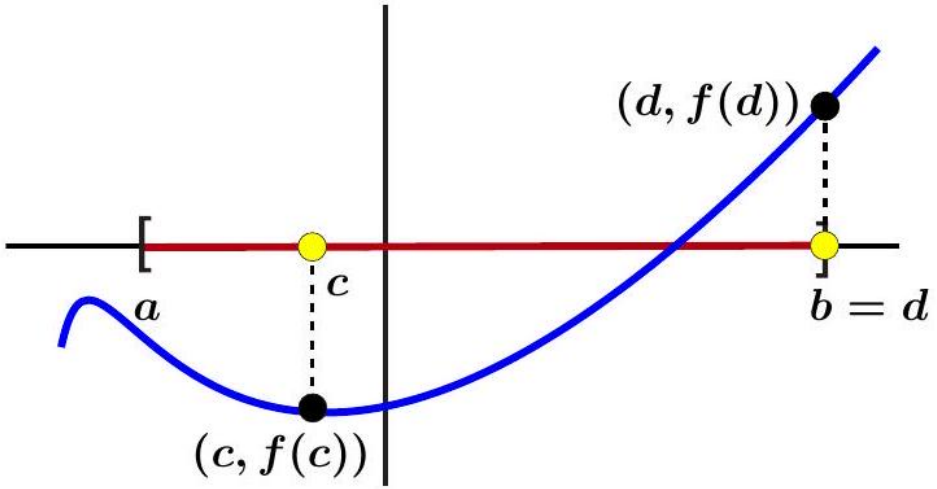
A point c in the domain of a function $f(x)$ is called a critical point for $f(x)$ if either

$$f'(c) = 0$$

or

$$f'(c) \text{ does not exist.}$$

Global Extrema and the EVT



Theorem 4.1.4 (Extreme Value Theorem (EVT))

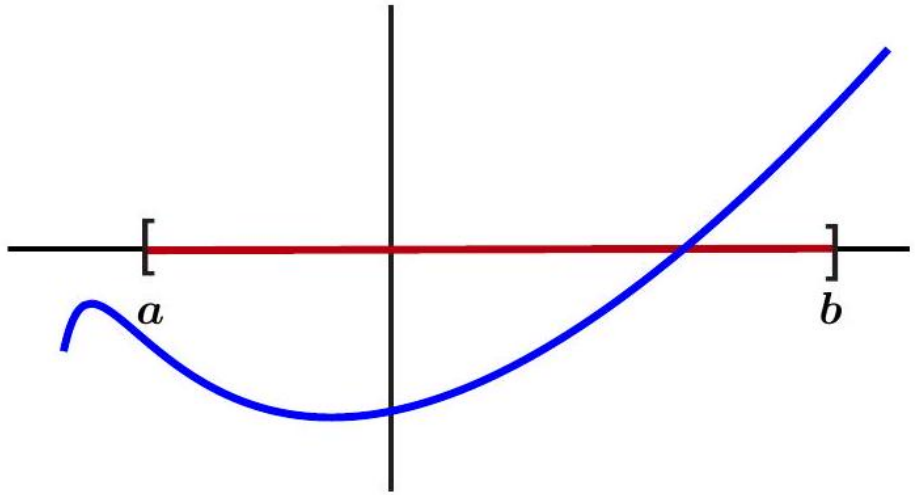
Assume that $f(x)$ is continuous on $[a, b]$. Then there exists $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d)$$

for every $x \in [a, b]$.

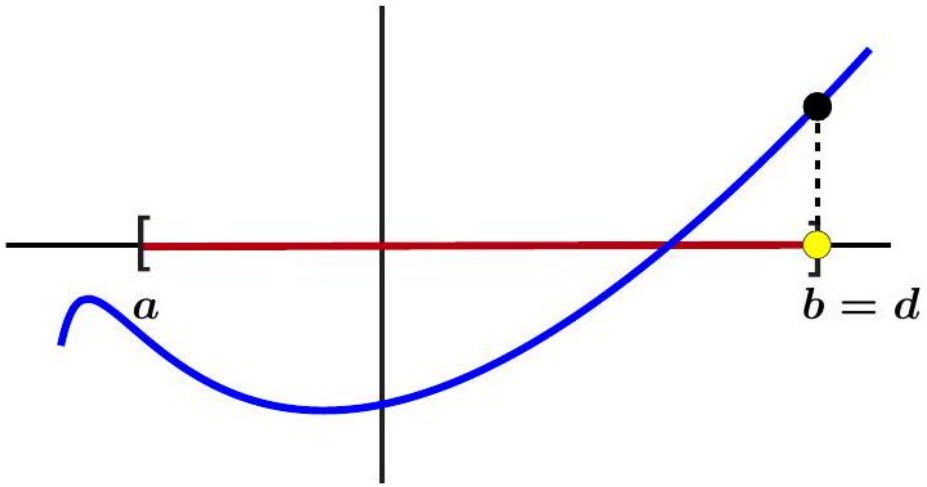
Question: How can we find c and d ?

Global Extrema and the EVT



Two Cases:

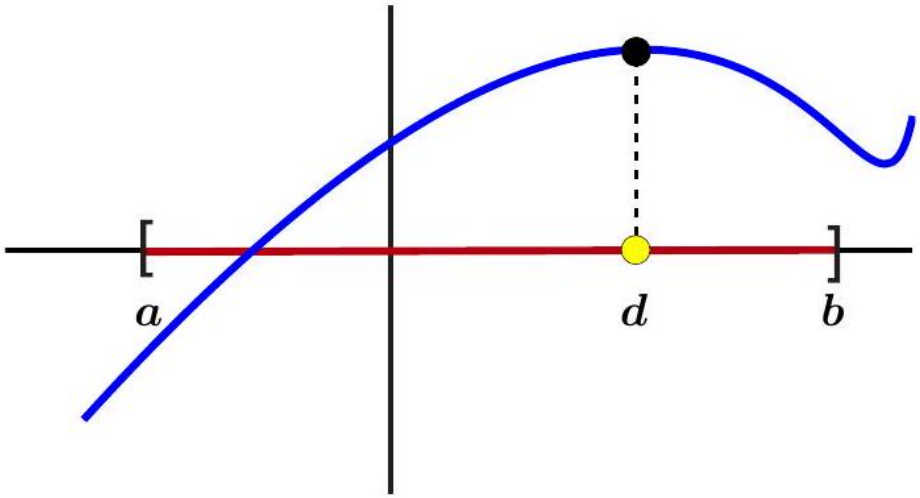
Global Extrema and the EVT



Two Cases:

1. d is an endpoint.

Global Extrema and the EVT



Two Cases:

1. d is an endpoint.
2. $d \in (a, b) \Rightarrow d$ is a local max $\Rightarrow d$ is a critical point.

Similarly for $\boldsymbol{c}$.

Finding Local Extrema

Question: How do we know if a critical point $x = c$ is actually a local maximum or a local minimum?

First Derivative Test

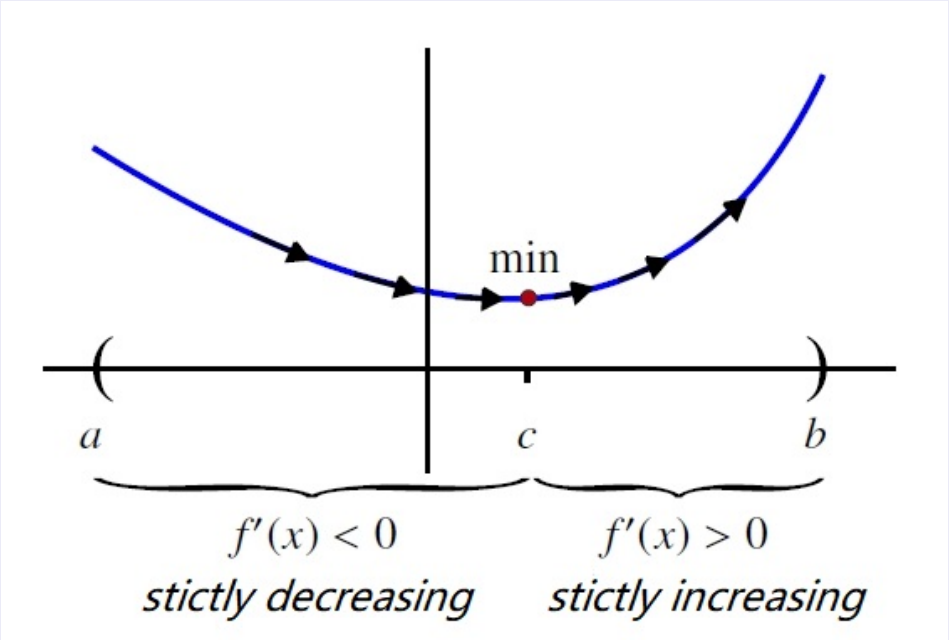
Definition 4.1.5 (Critical Point)

A point c in the domain of a function $f(x)$ is called a critical point if either $f'(c) = 0$ or $f'(c)$ does not exist.

Problem: Given a critical point c for $f(x)$, how can we determine if it is either a local maximum or a local minimum?

First Derivative Test

Theorem 4.1.6 (First Derivative Test)



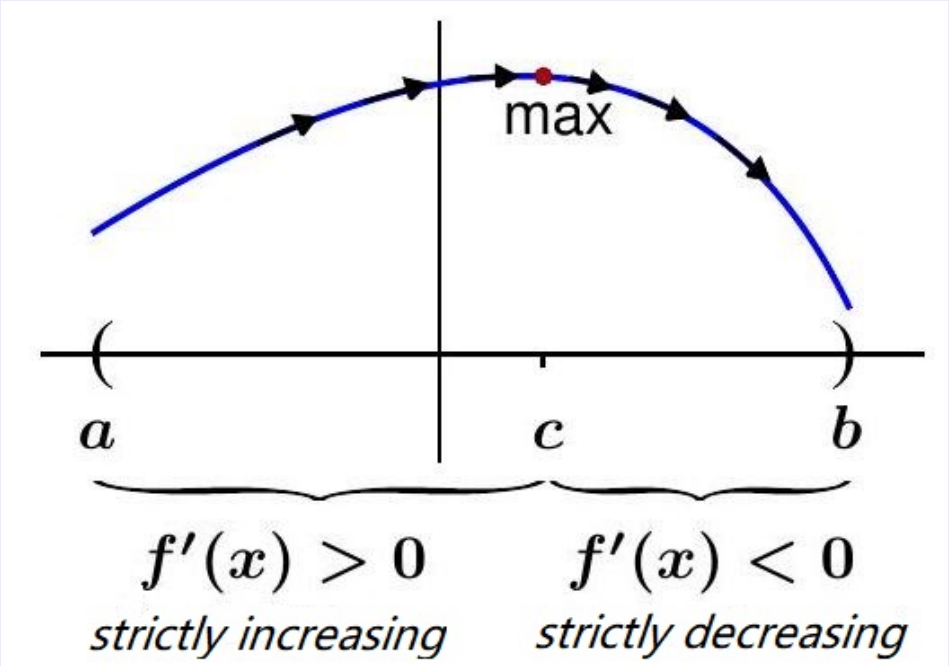
Assume that c is a critical point of $f(x)$, and $f(x)$ is continuous at c.
i) If there is an interval (a, b) containing c such that

$$\begin{aligned} & f'(x) < 0 \quad \text{for all } x \in (a, c) \\ \text{and} \quad & f'(x) > 0 \quad \text{for all } x \in (c, b) \end{aligned}$$

then $f(x)$ has a local minimum at $x = c$.

First Derivative Test

Theorem 4.1.6 (First Derivative Test) (Continue)



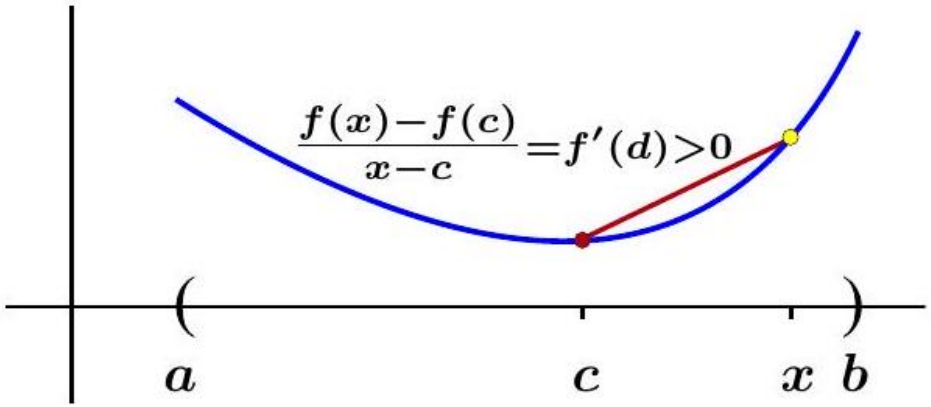
ii) If there is an interval (a, b) containing c such that

$$\begin{aligned} & f'(x) > 0 && \text{for all } x \in (a, c) \\ \text{and} \quad & f'(x) < 0 && \text{for all } x \in (c, b) \end{aligned}$$

then $f(x)$ has a local maximum at $x = c$

First Derivative Test

Proof of i)



Assume that $f'(x) < 0$ for all $x \in (a, c)$ and $f'(x) > 0$ for all $x \in (c, b)$.

i) Let $x \in (a, c)$. Then

$$\frac{f(x) - f(c)}{x - c} = f'(d) < 0 \Rightarrow f(x) - f(c) > 0$$

ii) Let $x \in (c, b)$. Then

$$\frac{f(x) - f(c)}{x - c} = f'(d) > 0 \Rightarrow f(x) - f(c) > 0$$

First Derivative Test

Example 4.1.7

Find all the critical points for

$$f(x) = \frac{x^3}{3} - x$$

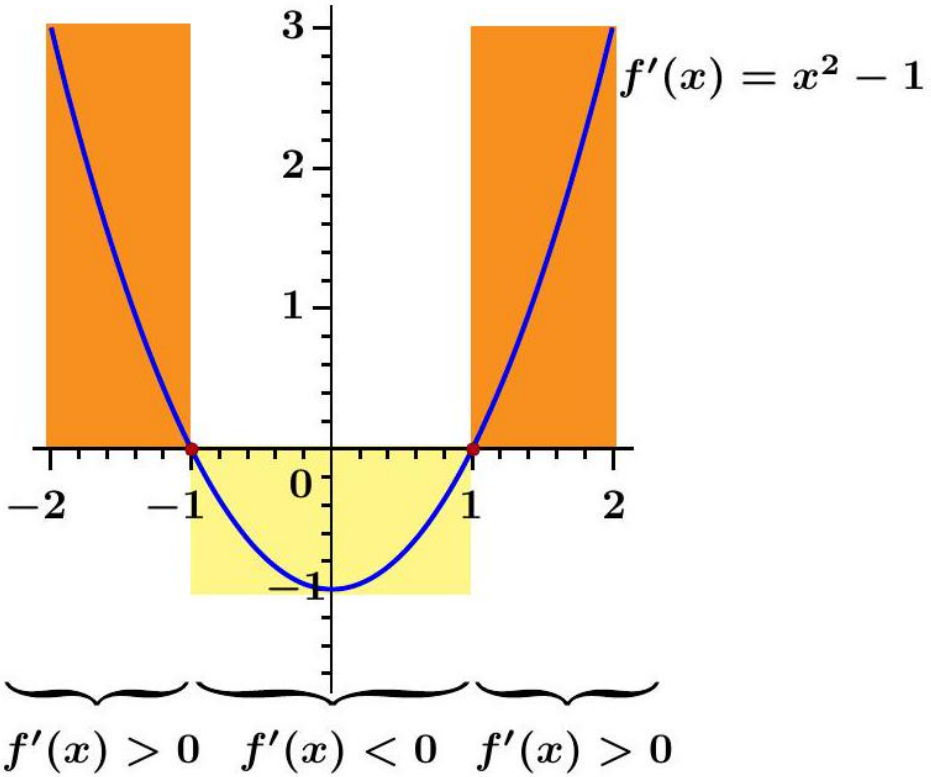
and determine if each is a local maximum, a local minimum, or neither.

Solution: We need to examine the first derivative:

$$f'(x) = x^2 - 1 = (x-1)(x+1)$$

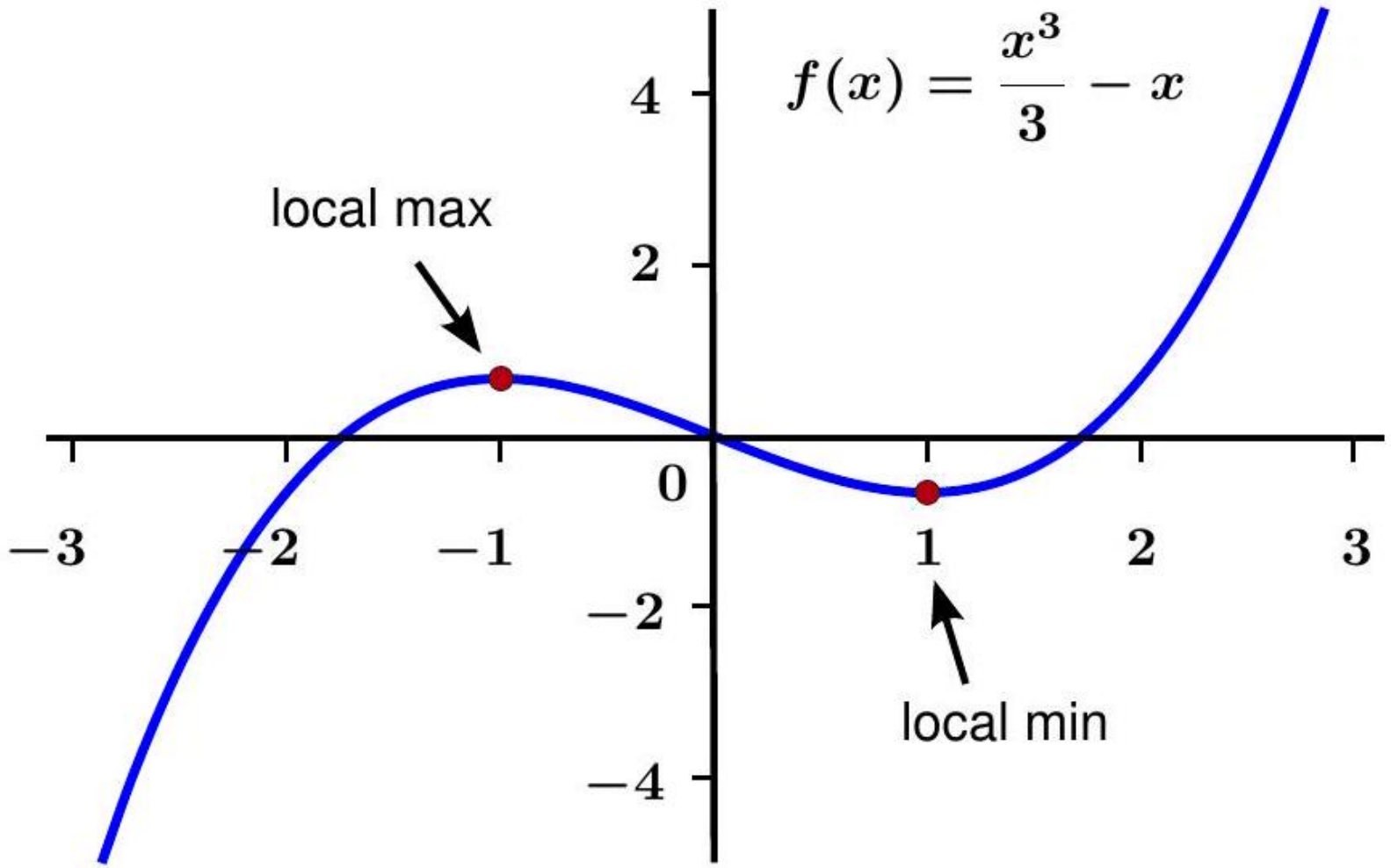
so the only critical points are $x = -1$ and $x = 1$.

First Derivative Test

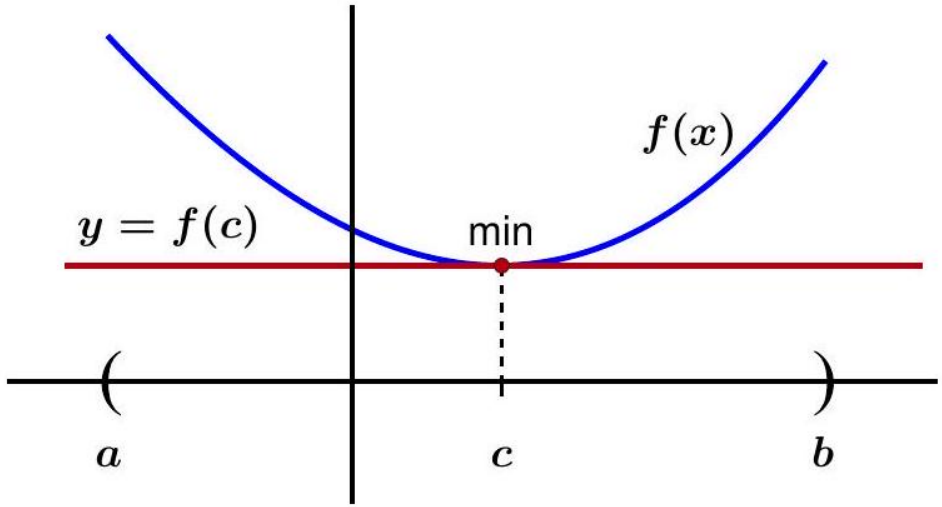


1. $f'(x)$ changes from positive to negative at $x = -1 \Rightarrow$ local maximum.
2. $f'(x)$ changes from negative to positive at $x = 1 \Rightarrow$ local minimum.

First Derivative Test



Critical Points and Concavity



Problem: Given a critical point c for $f(x)$ with $f'(c) = 0$, does concavity help us determine whether c is either a local maximum or a local minimum?

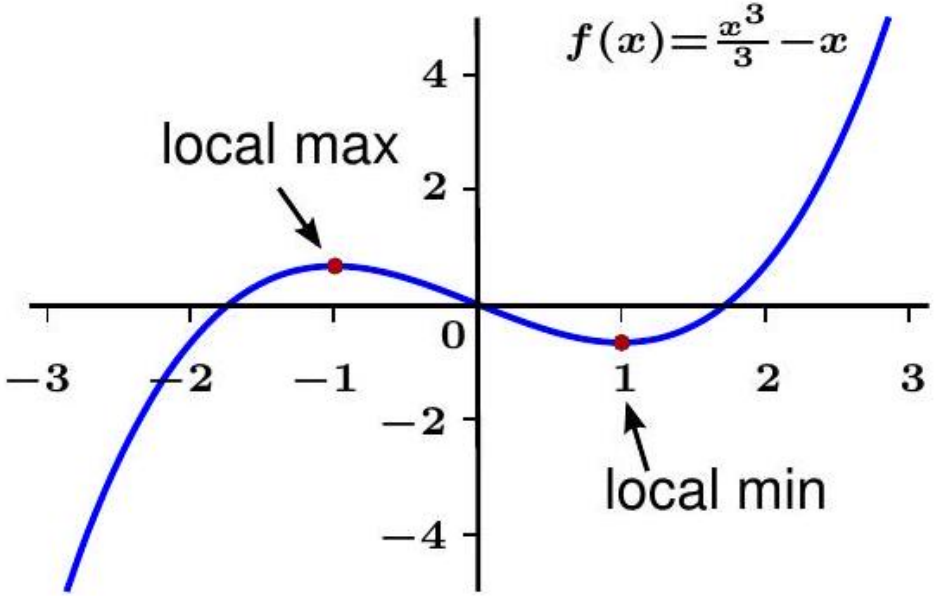
1. If $f(x)$ is concave on an interval containing $x = c$, then $(c, f(c))$ should be a local maximum.
2. If $f(x)$ is convex on an interval containing $x = c$, then $(c, f(c))$ should be a local minimum.

The Second Derivative Test:

Theorem 4.1.8 (Second Derivative Test)

Assume that $f'(c) = 0$ *and that* $f''(x)$ *is continuous at* $x = c$.
i) If $f''(c) < 0$, *then* $f(x)$ *has a local maximum at* c.
ii) If $f''(c) > 0$, *then* $f(x)$ *has a local minimum at* c.

The Second Derivative Test



Example: Let

$$f(x) = \frac{x^3}{3} - x$$

Then

$$f'(x) = x^2 - 1$$

so the critical points are $x = -1$ and $x = 1$.
We have $f''(x) = 2x$. Since
i) $f''(-1) < 0$, then $f(x)$ has a local maximum at $x = -1$. ii) $f''(1) > 0$, then $f(x)$ has a local minimum at $x = 1$.

Extreme Value Theorem

Theorem 4.1.9 (The Extreme Value Theorem (EVT))

Suppose that $f(x)$ *is continuous on* $[a, b]$. *There exists* c_1 *and* $c_2 \in [a, b]$ *such that*

$$f(c_1) \leq f(x) \leq f(c_2)$$

for all $x \in [a, b]$.

Extreme Value Theorem

Observation: The EVT ensures that a continuous function $f(x)$ defined on a closed interval $[a, b]$ achieves its global maximum and global minimum on $[a, b]$, but it does not tell us how to find these values.

Important Fact: Assume that $f(x)$ has either a global maximum or global minimum at $c \in [a, b]$. Then either

1. c is an endpoint of the interval $[a, b] \Rightarrow c = a$ or $c = b$

   or

2. c is not an endpoint of the interval $[a, b] \Rightarrow c \in (a, b)$ is a local maximum or local minimum $\Rightarrow c$ is a critical point.

Global Maxima and Global Minima

Summary [Finding the Global Maximum and Global Minimum]

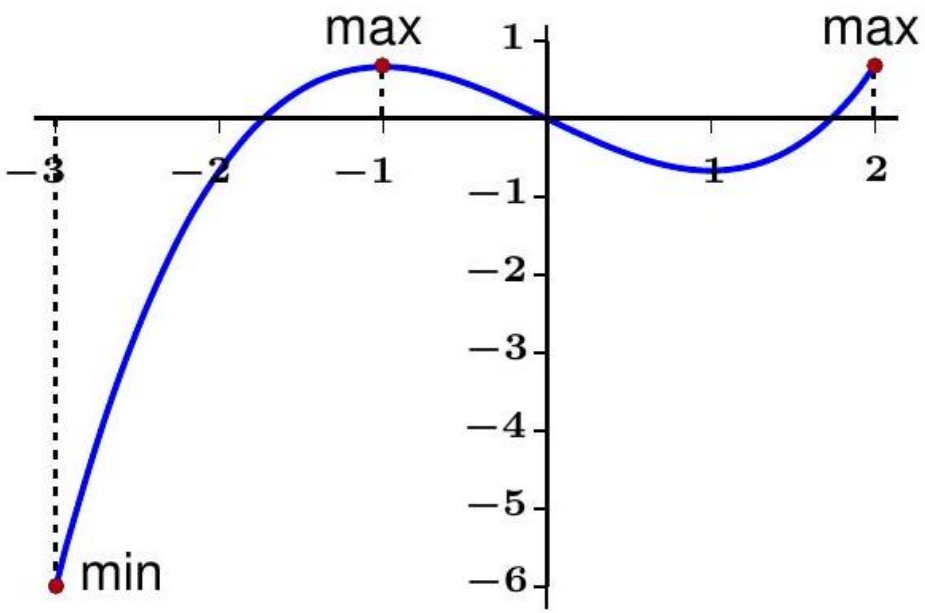
To find the maximum and minimum for a continuous function $f(x)$ on $[a, b]$:
Step 1: Evaluate $f(a)$ and $f(b)$.

Step 2: Find all critical points c in (a, b) such that $f'(c) = 0$ and $f'(c)$ does not exist, where applicable.

Step 3: Evaluate the function at each of the critical points.

Step 4: The global maximum is at the point that produces the largest value from Steps 1 and 3. The global minimum is at the point that produces the smallest value from Steps 1 and 3.

Global Maxima and Global Minima



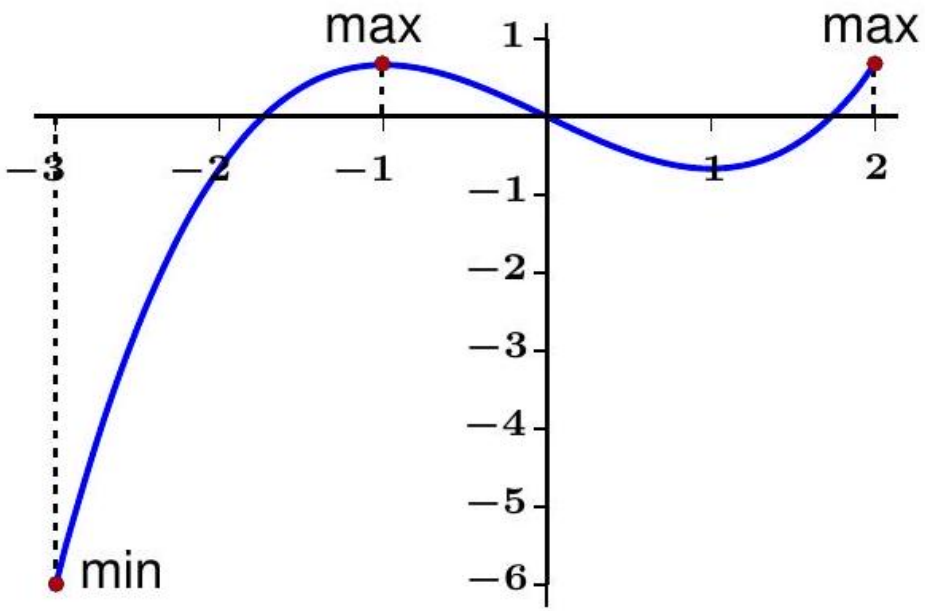
Example 4.1.10

Let

$$f(x) = \frac{x^3}{3} - x$$

Find the maximum and minimum values of $f(x)$ on $[-3, 2]$.

Global Maxima and Global Minima



Solution:

Step 1: Evaluate endpoints: $f(-3) = -6$ and $f(2) = \frac{2}{3}$.

Step 2: Since $f'(x) = x^2 - 1$ the critical points are $x = \pm 1$.

Step 3: $f(-1) = \frac{2}{3}$ and $f(1) = -\frac{2}{3}$.

Step 4: The global maximum is at both $x = -1$ and $x = 2$ where the maximum value is $\frac{2}{3}$. The global minimum is at $x = -3$ where the minimum value is -6 .

Section 2

Increasing Function Theorem

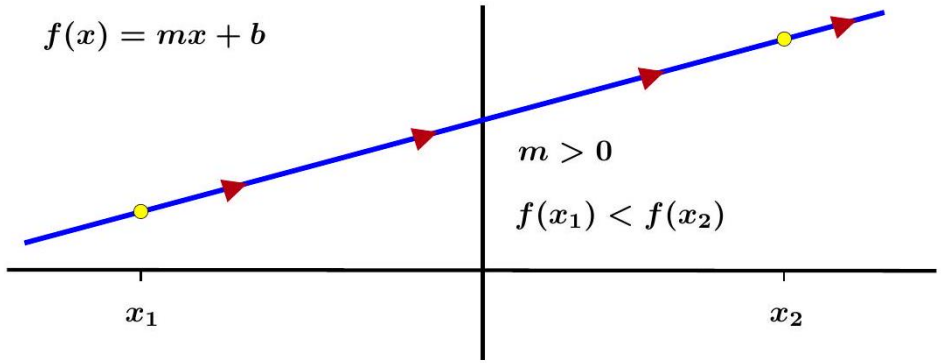
Functions

Definition 4.2.1 (Increasing and Decreasing Functions)

Suppose that $f(x)$ is defined on an interval I.
i) We say that $f(x)$ is strictly increasing on I if $f(x_1) < f(x_2)$ for all $x_1, x_2 \in I$ with $x_1 < x_2$.
ii) We say that $f(x)$ is strictly decreasing on I if $f(x_1) > f(x_2)$ for all $x_1, x_2 \in I$ with $x_1 < x_2$.
iii) We say that $f(x)$ is increasing on I if $f(x_1) \leq f(x_2)$ for all $x_1, x_2 \in I$ with $x_1 < x_2$.
iv) We say that $f(x)$ is decreasing on I if $f(x_1) \geq f(x_2)$ for all $x_1, x_2 \in I$ with $x_1 < x_2$.
Such functions are said to be monotonic on $\boldsymbol{I}$.

Question: How can we determine if a function $f(x)$ is either increasing or decreasing on an interval $\boldsymbol{I}$?

Increasing Function Theorem



Observation: Assume that

$$f(x) = mx + b$$

If $m > 0$, the graph of the function slopes upward as we move from left to right. In other words, if $x_1 < x_2$, then

$$f(x_1) = mx_1 + b < mx_2 + b = f(x_2)$$

Note: $f'(x) = m > 0$ for all $x \in \mathbb{R}$.

Increasing Function Theorem

$f'(x) > 0$

$\boldsymbol{f(x)}$ always strictly increasing?

Question: If $f(x)$ is such that $f'(x) > 0$ for all $x \in I$, is $f(x)$ strictly increasing on I?

Increasing Function Theorem

Theorem 4.2.2 (The Increasing/Decreasing Function Theorem)

i) Let I be an interval and assume that $f'(x) > 0$ for all $x \in I$. If $x_1 < x_2$ are two points in I, then

$$f(x_1) < f(x_2)$$

That is, $\boldsymbol{f(x)}$ is strictly increasing on $\boldsymbol{I}$.

ii) Let I be an interval and assume that $f'(x) \geq 0$ for all $x \in I$. If $x_1 < x_2$ are two points in I, then

$$f(x_1) \leq f(x_2)$$

That is, $\boldsymbol{f(x)}$ is increasing on $\boldsymbol{I}$.

Increasing Function Theorem

Theorem 4.2.2 (The Increasing/Decreasing Function Theorem)

iii) Let I be an interval and assume that $f'(x) < 0$ for all $x \in I$. If $x_1 < x_2$ are two points in I, then

$$f(x_1) > f(x_2)$$

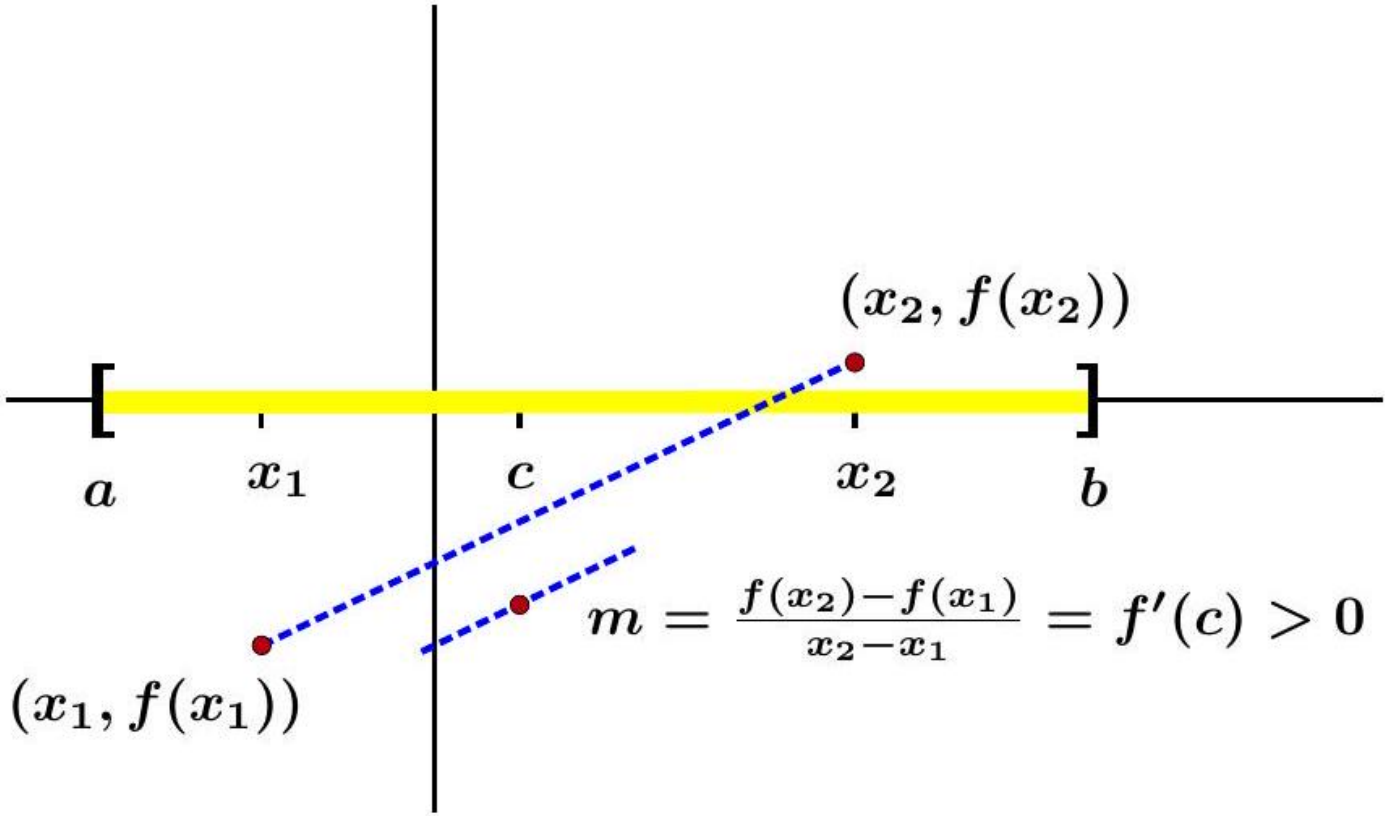
That is, $\boldsymbol{f(x)}$ is strictly decreasing on $\boldsymbol{I}$.

iv) Let I be an interval and assume that $f'(x) \leq 0$ for all $x \in I$. If $x_1 < x_2$ are two points in I, then

$$f(x_1) \geq f(x_2)$$

That is, $f(x)$ is decreasing on I.

Increasing Function Theorem



i) Let I be an interval and assume that $f'(x) > 0$ for all $x \in I$. If $x_1 < x_2$ are two points in I, then

$$f(x_1) < f(x_2)$$

Increasing Function Theorem

Proof of i): Assume that $f'(x) > 0$ for all $x \in I$. Let $x_1, x_2 \in I$ with $x_1 < x_2$. By the MVT there exists $c \in (x_1, x_2)$ with

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) > 0$$

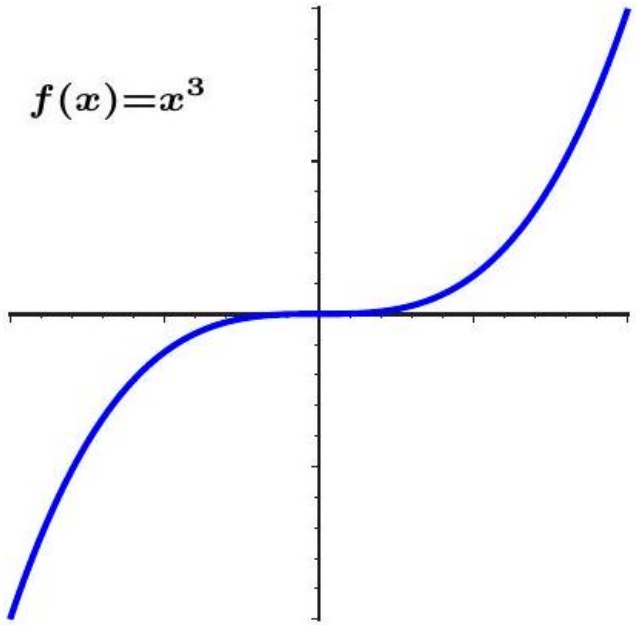
Since $x_2 - x_1 > 0$, we have

$$f(x_2) - f(x_1) > 0$$

and hence that

$$f(x_2) > f(x_1)$$

Increasing Function Theorem



Question: If $f(x)$ is strictly increasing on an interval I and differentiable on I, then must $f'(x) > 0$ for all $x \in I$?

Solution: Let $f(x) = x^3$. Since $f'(x) = 3x^2$, we have

$$f'(0) = 0$$

but $f(x)$ is strictly increasing on all of $\mathbb{R}$.

Increasing Function Theorem

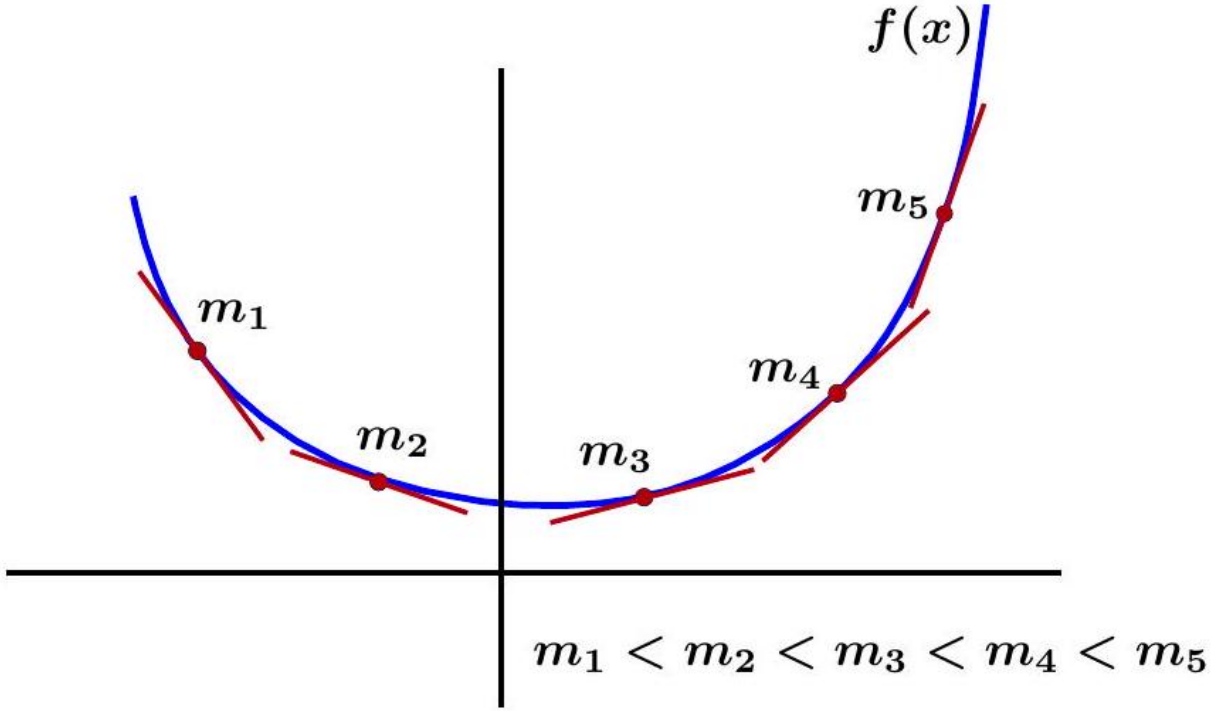
Question:

1. Is the function $f(x) = x^3$ strictly increasing on $[0, 1]$? Yes!
2. If $f(x)$ is everywhere differentiable and if $f'(c) > 0$, does this mean that there is an open interval (a, b) containing c on which $f(x)$ is increasing? No!

Section 3

Convexity and Concavity of Functions

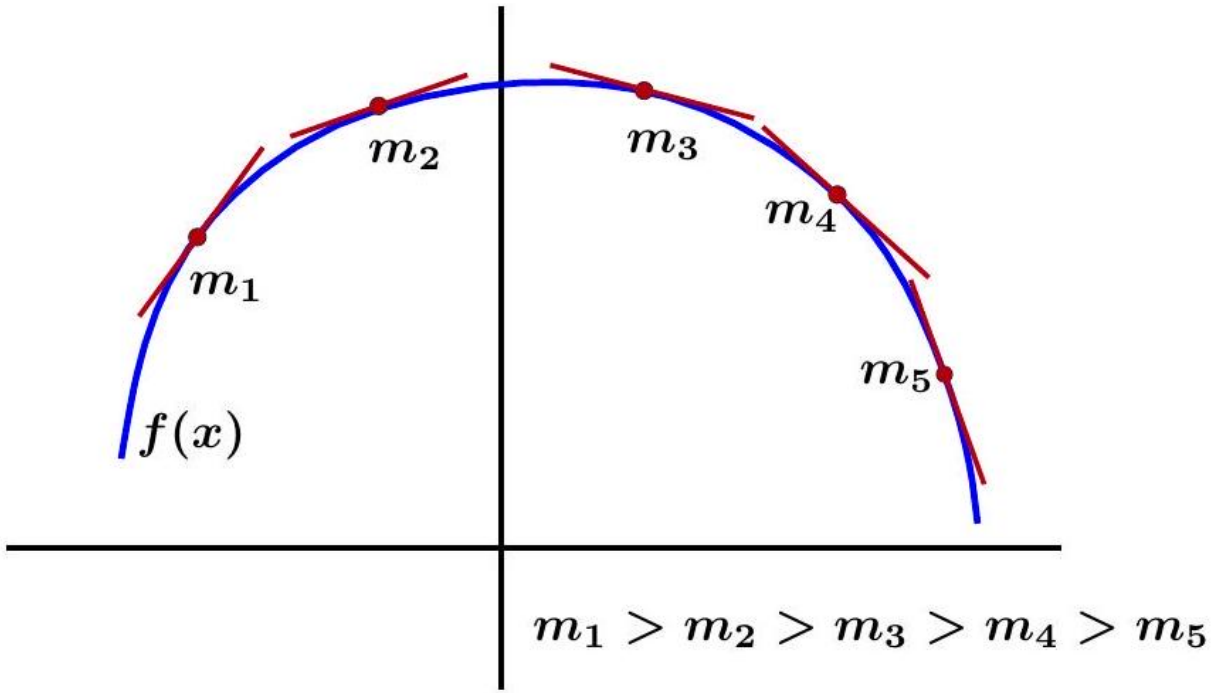
Convexity



Question: If $f''(x) > 0$ on some interval I, what can we say about the shape of the graph of the function $f(x)$?

Observation: $f'(x)$ is strictly increasing so the tangent lines to $f(x)$ rotate counter-clockwise $\Rightarrow f(x)$ is convex.

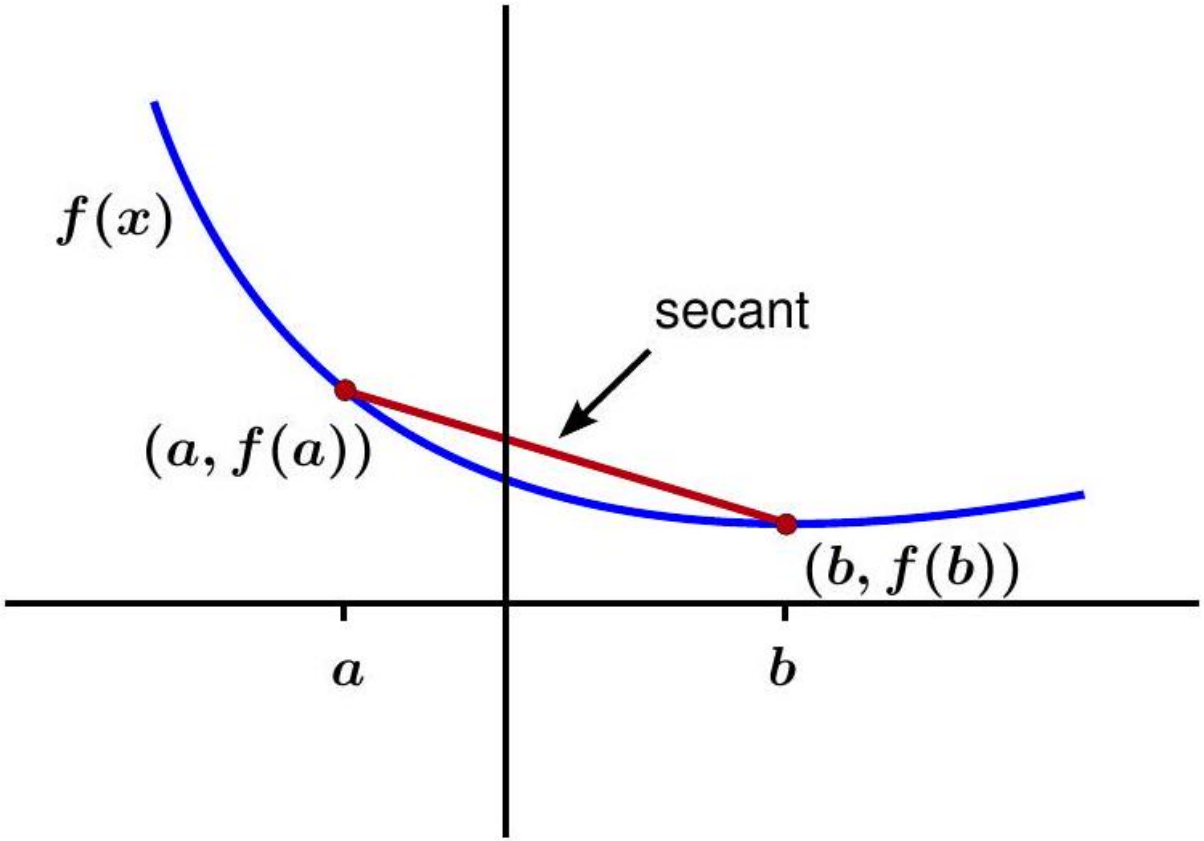
Concavity



Question: What if $f''(x) < 0$ on I ?

Observation: $f'(x)$ is strictly decreasing so tangent lines to $f(x)$ rotate clockwise $\Rightarrow f(x)$ is concave.

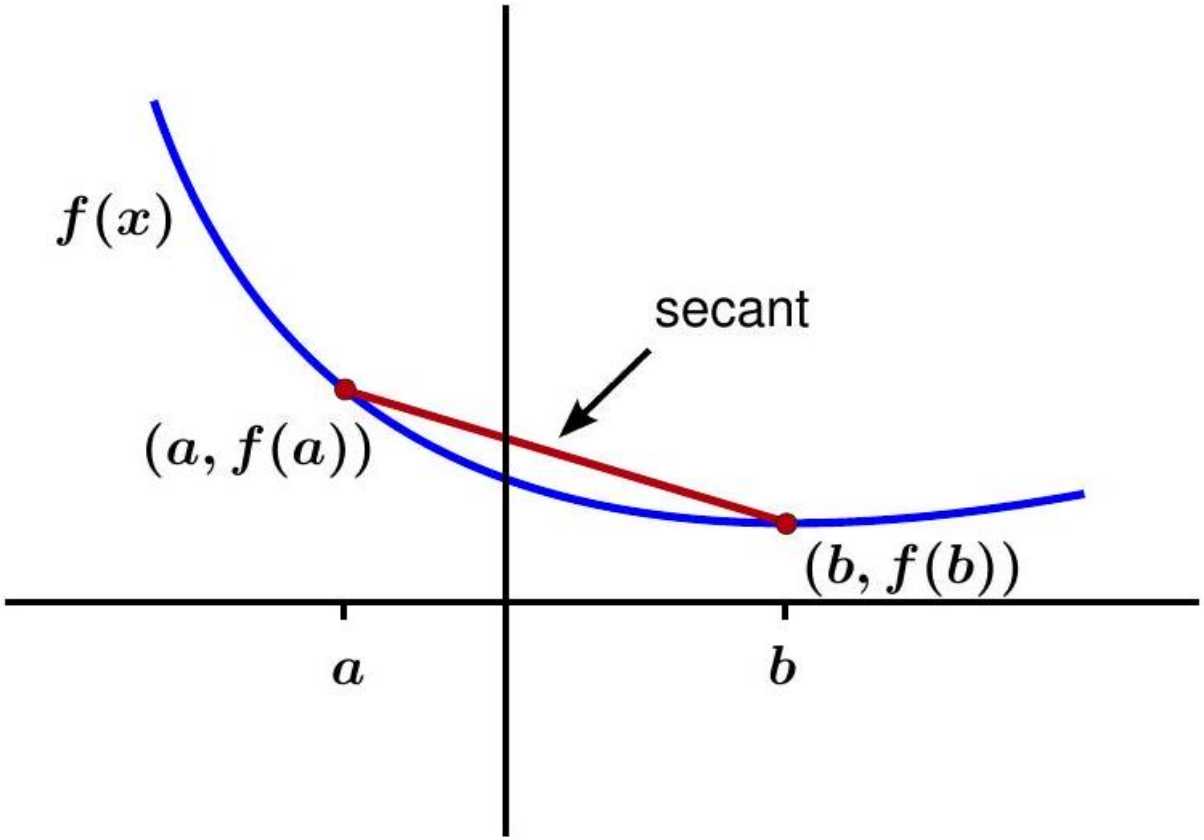
Convexity



Question: How can we quantify convexity?

Observation: The secant line lies above the graph of $f(x)$.

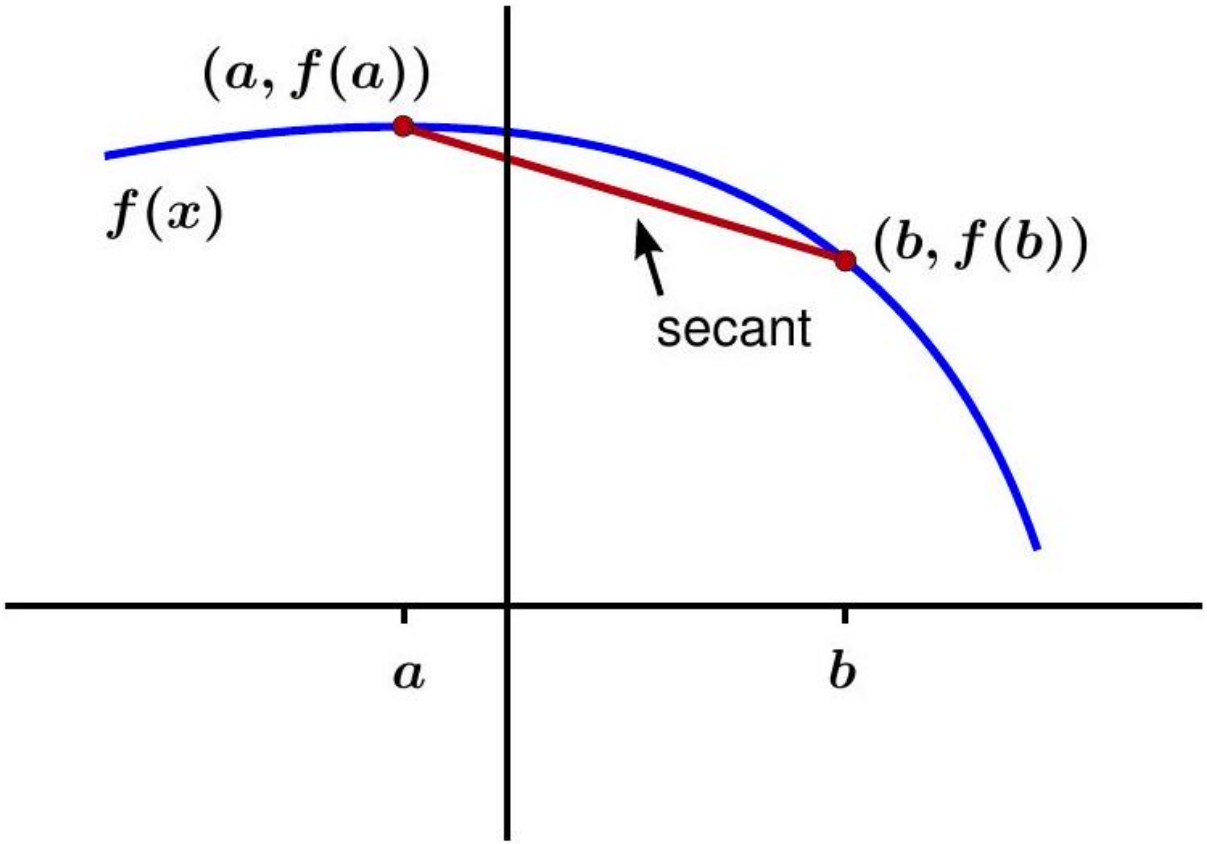
Convexity



Definition 4.3.1 (Convexity)

The graph of $f(x)$ is convex on an interval I if for every pair of points a and b in I, the secant line joining $(a, f(a))$ and $(b, f(b))$ lies above the graph of $f(x)$.

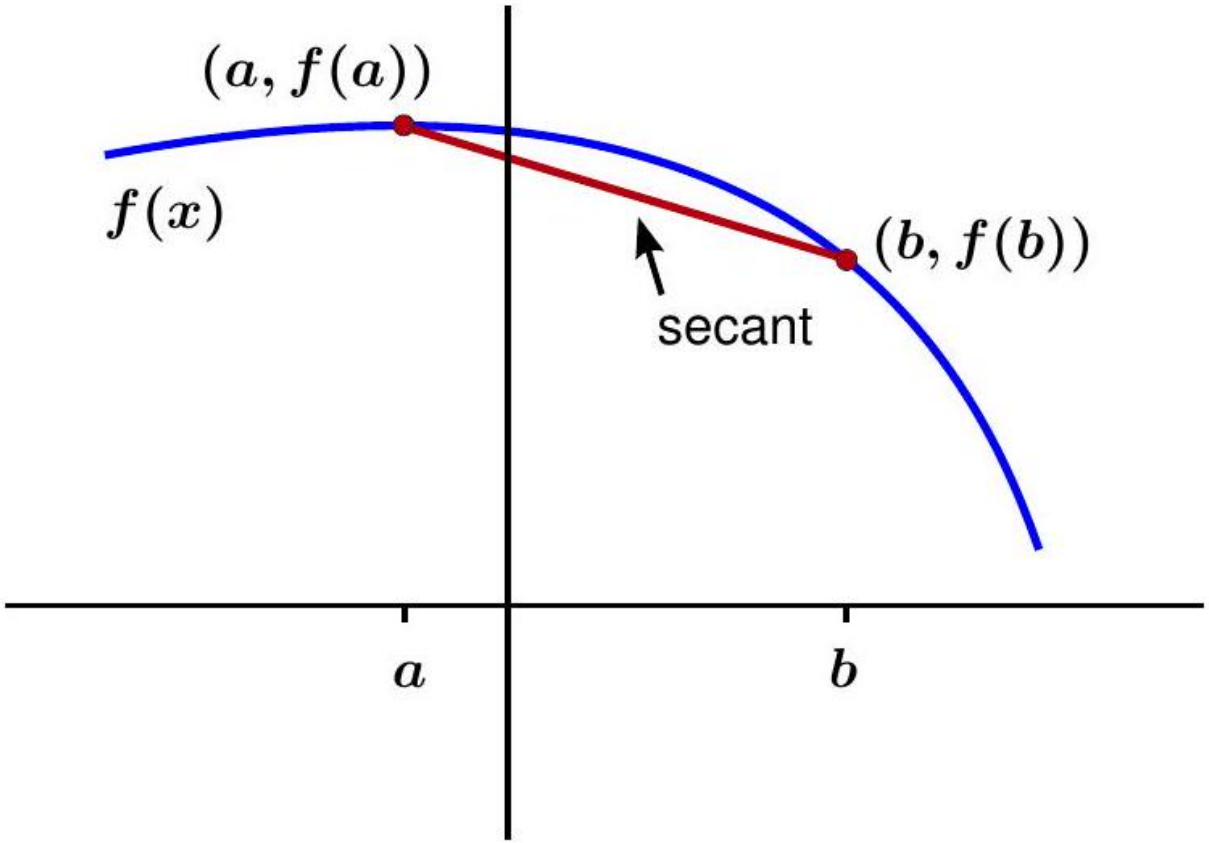
Concavity



Question: How can we quantify concavity?

Observation: The secant line lies below the graph of $f(x)$.

Concavity



Definition 4.3.2 (Concavity)

The graph of $f(x)$ is concave on an interval I if for every pair of points a and b in I, the secant line joining $(a, f(a))$ and $(b, f(b))$ lies below the graph of $f(x)$.

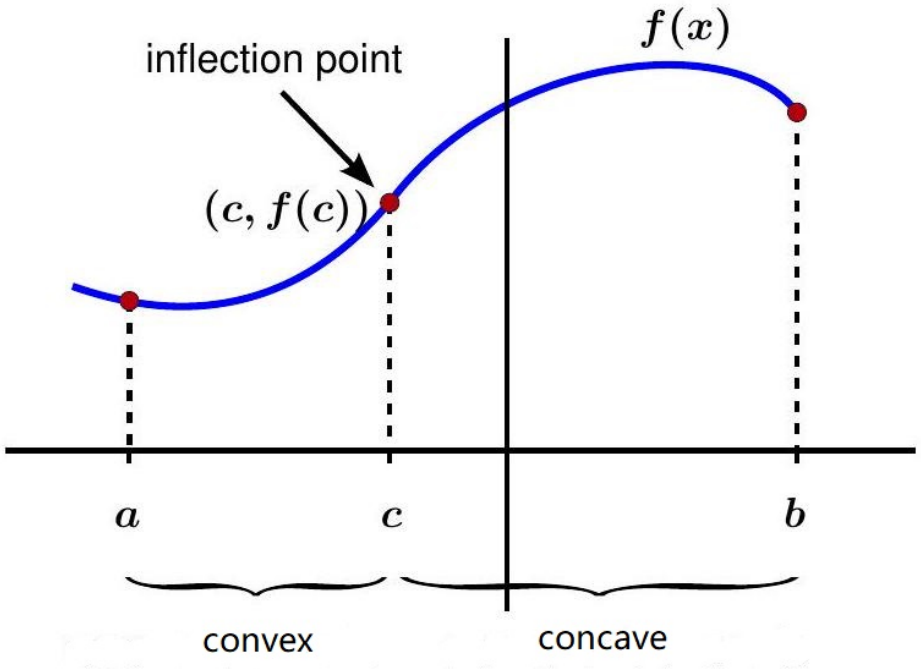
Second Derivative Test for Convexity and Concavity

Theorem 4.3.3 (Second Derivative Test for Convexity and Concavity)

i) If $f''(x) > 0$ for each x in an interval I, then the graph of $f(x)$ is convex on I.

ii) If $f''(x) < 0$ for each x in an interval I, then the graph of $f(x)$ is concave on I.

Inflection Points



Example:

1. $f(x)$ is convex on $[a, c]$.
2. $f(x)$ is concave on $[c, b]$.
3. $f(x)$ changes from convexity to concavity at $x = c$. The point $(c, f(c))$ is called an inflection point.

Inflection Points

Definition 4.3.4 (Inflection Point)

A point c is called an inflection point for the function $f(x)$ if
i) $f(x)$ is continuous at $x = c$, and ii) the convexity or concavity of $f(x)$ changes at $x = c$.

Important Note:

1. Typically an inflection point $x = c$ would occur when the second derivative changes from positive to negative, or vice versa.
2. If $f''(x)$ is continuous, the Intermediate Value Theorem requires that $f''(c) = 0$.

Inflection Points

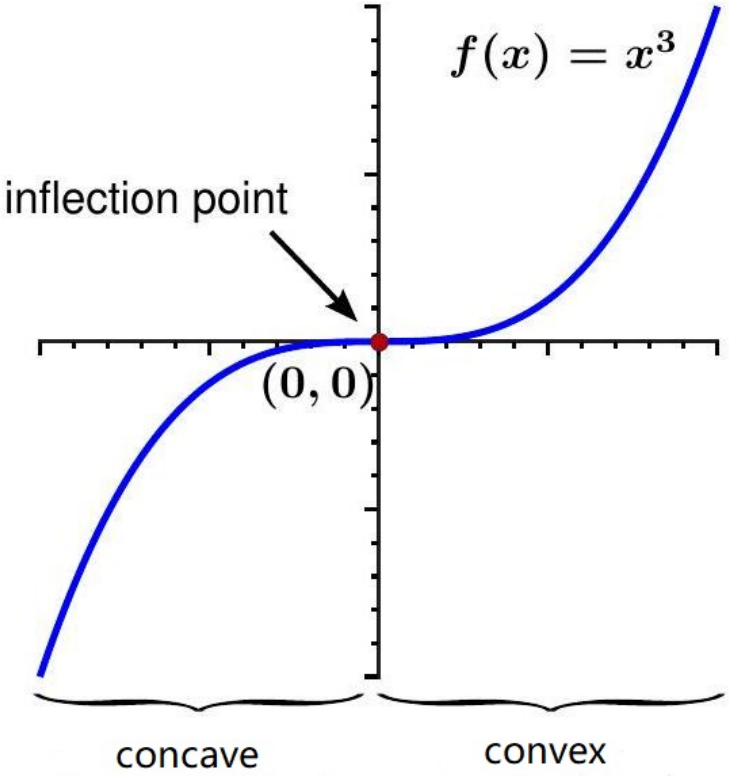
Theorem 4.3.5 (Test for Inflection Points)

If $f''(x)$ is continuous at $x = c$ and $(c, f(c))$ is an inflection point for $f(x)$, then

$$f''(c) = 0$$

WARNING: This theorem shows us how to locate candidates for inflection points. However, $f''(c) = 0$ does not mean that an inflection point always occurs when $x = c$.

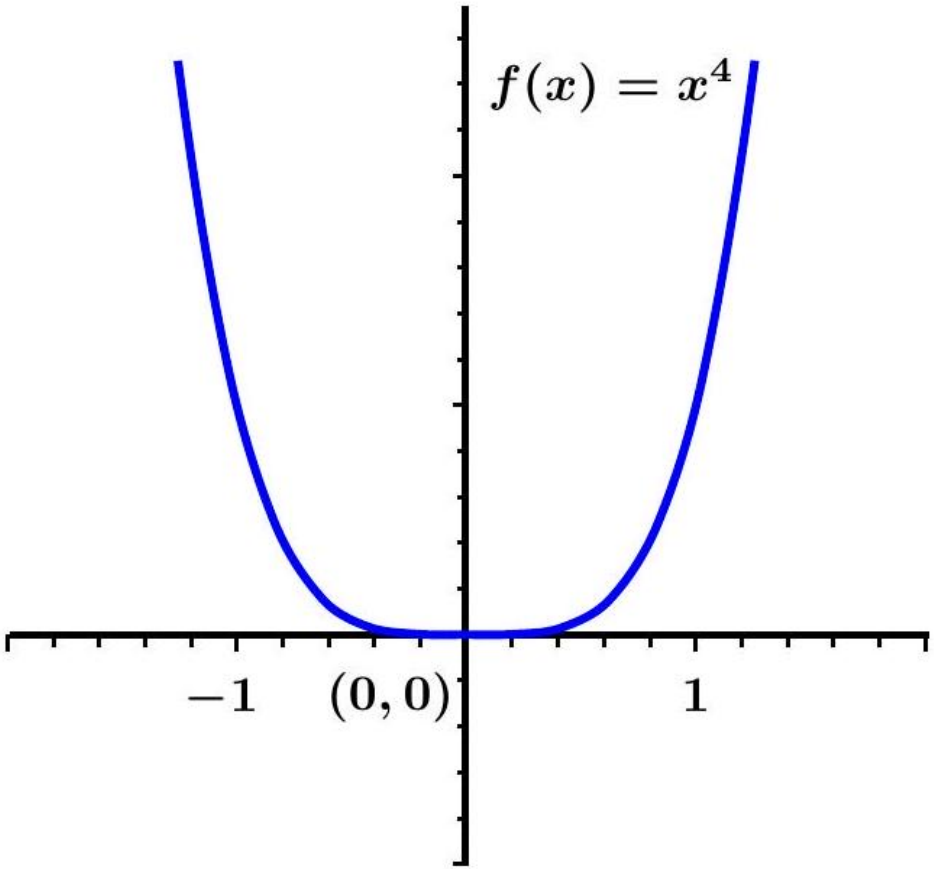
Inflection Points



Example: Let $f(x) = x^3$. Then $f''(x) = 6x$.

1. $f''(x) < 0$ if $x < 0$ so $f(x)$ is concave on $(-\infty, 0]$.
2. $f''(x) > 0$ if $x > 0$ so $f(x)$ is convex on $[0, \infty)$.
3. The function has a point of inflection at $\boldsymbol{x = 0}$.

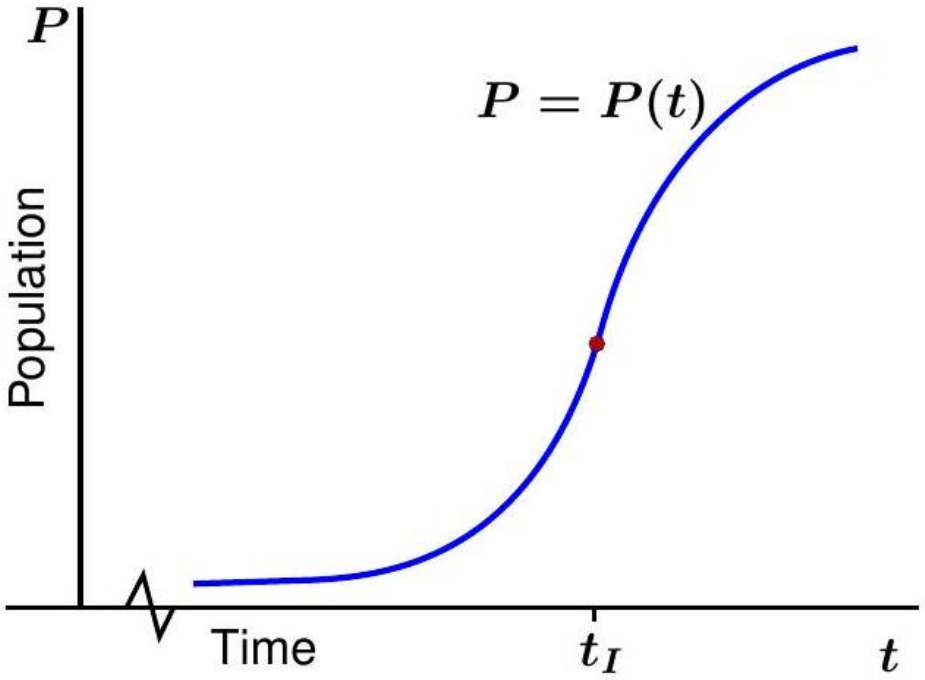
Inflection Points



Example: Let $f(x) = x^4$. Then $f''(x) = 12x^2$.

1. $f''(x) > 0$ for all $x \neq 0$ so $f(x)$ is convex on $\mathbb{R}$.
2. $f''(0) = 0$ but the function does not have a point of inflection at $x = 0$.

Inflection Points



Example: The diagram represents the graph of the population P of a particular bacteria over time t in a restricted environment.

Important Fact: The point of inflection occurs at the time t_I when the growth rate of the bacteria is at its highest value.

Section 4

L'Hôpital's Rule

L'Hôpital's Rule

Recall: If $h(x) = \frac{f(x)}{g(x)}$ and if

$$\lim_{x \to a} f(x) = 0 = \lim_{x \to a} g(x)$$

then we do not know whether or not $\lim_{x \to a} h(x)$ exists.

For this reason, we call such a situation an indeterminate form of type $\frac{0}{0}$.

Similarly, if

$$\lim_{x \to a} f(x) = \pm\infty = \lim_{x \to a} g(x)$$

we would not be able to determine immediately if the limit of the quotient exists.

We call this situation an indeterminate form of type $\frac{\infty}{\infty}$.

Note: L'Hôpital's Rule gives us a means to evaluate such limts.

Observation: Let $h(x) = \frac{f(x)}{g(x)}$ and $\lim_{x\to a} f(x) = 0 = \lim_{x\to a} g(x)$.

Assume that $f(x)$ and $g(x)$ have continuous derivatives with $g'(a) \neq 0$. We know that for $\boldsymbol{x}$ near $\boldsymbol{a}$ we have that

$$\frac{f(x)}{g(x)} \cong \frac{f(a) + f'(a)(x-a)}{g(a) + g'(a)(x-a)} = \frac{f'(a)}{g'(a)}$$

since $f(a) = 0 = g(a)$.

This might lead us to guess that if $\lim_{x\to a} \frac{f(x)}{g(x)}$ exists, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)} \tag{1}$$

Moreover, since $f'(x)$ and $g'(x)$ are continuous with $g'(a) \neq 0$, we also have

$$\frac{f'(a)}{g'(a)} = \lim_{x\to a} \frac{f'(x)}{g'(x)} \tag{2}$$

Combining (1) and (2) gives us

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

L'Hospital's Rule

Theorem 4.4.1 (L'Hôpital's Rule)

Assume that $f'(x)$ and $g'(x)$ exist near $x = a, g'(x) \neq 0$ near $x = a$ except possibly at $x = a$, and that $\lim_{x\to a} \frac{f(x)}{g(x)}$ is an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

provided that the latter limit exists (or is ∞ or $-\infty$).

Moreover, this rule remains valid for one-sided limits and for limits at $\pm\infty$.

Note: The proof of L'Hôpital's Rule uses an upgraded version of the MVT.

L'Hôpital's Rule

Example: Evaluate

$$\lim_{x\to 0} \frac{e^x - 1}{x}$$

Solution: Let $f(x) = e^x - 1$ and $g(x) = x$. Then

$$\lim_{x\to 0} e^x - 1 = e^0 - 1 = 0 = \lim_{x\to 0} x \Rightarrow \text{ type } \frac{0}{0}$$

Since $f'(x) = e^x$ and $g'(x) = 1$, by L'Hôpital's Rule

$$\lim_{x\to 0} \frac{e^x - 1}{x} = \lim_{x\to 0} \frac{e^x}{1} = 1$$

Note:

$$\lim_{x\to 0} \frac{e^x - 1}{x}$$

is the derivative of $f(x) = e^x$ at $x = 0$.

L'Hôpital's Rule

Example: Evaluate

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$$

Solution: Let $f(x) = e^x - 1 - x$ and $g(x) = x^2$. Then

$$\lim_{x\to 0} e^x - 1 - x = 0 = \lim_{x\to 0} x^2 \Rightarrow \text{ type } \frac{0}{0}$$

Then $f'(x) = e^x - 1$ and $g'(x) = 2x$ and

$$\lim_{x\to 0} e^x - 1 = 0 = \lim_{x\to 0} 2x \Rightarrow \lim_{x\to 0} \frac{f'(x)}{g'(x)} \text{ is type } \frac{0}{0}$$

Let $F(x) = f'(x) = e^x - 1$ and $G(x) = g'(x) = 2x$, then $F'(x) = e^x$ and $G'(x) = 2$ so

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x^2} = \lim_{x\to 0} \frac{f'(x)}{g'(x)} = \lim_{x\to 0} \frac{F(x)}{G(x)} = \lim_{x\to 0} \frac{e^x}{2} = \frac{1}{2}$$

L'Hôpital's Rule

Example: Evaluate

$$\lim_{x \to 0^+} \frac{e^x - 1}{x^2}$$

Solution: Let $f(x) = e^x - 1$ and $g(x) = x^2$. Then $f'(x) = e^x$ and $g'(x) = 2x$, so we have

$$\lim_{x \to 0^+} \frac{e^x - 1}{x^2} = \lim_{x \to 0^+} \frac{e^x}{2x}$$

But

$$\lim_{x \to 0^+} 2x = 0$$

so by applying L'Hôpital's Rule again we get

$$\lim_{x \to 0^+} \frac{e^x - 1}{x^2} = \lim_{x \to 0^+} \frac{e^x}{2x} = \lim_{x \to 0^+} \frac{e^x}{2} = \frac{1}{2}$$

Warning: This is wrong since $\lim_{x \to 0^+} e^x = 1 \neq 0$ so

$$\lim_{x \to 0^+} \frac{e^x - 1}{x^2} = \lim_{x \to 0^+} \frac{e^x}{2x} = \infty$$

L'Hôpital's Rule

Example: Evaluate

$$\lim_{x\to\infty} \frac{\ln(x)}{x}$$

Solution: Let $f(x) = \ln(x)$ and $g(x) = x$. Then

$$\lim_{x\to\infty} f(x) = \infty = \lim_{x\to\infty} g(x)$$

so this is an indeterminate form of the type $\frac{\infty}{\infty}$.
Differentiating $f(x)$ and $g(x)$ gives us $f'(x) = \frac{1}{x}$ and $g'(x) = 1$. Therefore,

$$\begin{aligned}\lim_{x\to\infty} \frac{f'(x)}{g'(x)} &= \lim_{x\to\infty} \frac{\frac{1}{x}}{1} \\ &= \lim_{x\to\infty} \frac{1}{x} \\ &= 0\end{aligned}$$

Thus

$$\lim_{x\to\infty} \frac{\ln(x)}{x} = 0$$

L'Hôpital's Rule

Remark: Up until now we have dealt with two types of indeterminate forms which we have denoted by $\frac{0}{0}$ and $\frac{\infty}{\infty}$. There are five more standard indeterminate forms which we will denote by

$$0 \cdot \infty, \quad \infty - \infty, \quad 1^{\infty}, \quad \infty^0, \quad \text{and} \quad 0^0$$

For example, an indeterminate form of type $0 \cdot \infty$ arises from the function $h(x) = f(x)g(x)$ when

$$\lim_{x \to a} f(x) = 0$$

and

$$\lim_{x \to a} g(x) = \infty$$

Similarly, the function $(g(x))^{f(x)}$ would produce an indeterminate form of type ∞^0.

Note: All of the above forms can be converted to forms of the type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

L'Hôpital's Rule

Example: Evaluate

$$\lim_{x \to 0+} x \ln(x)$$

Solution: This is an indeterminate form of type $\mathbf{0 \cdot \infty}$ since

$$\lim_{x \to 0^+} x = 0$$

and

$$\lim_{x \to 0^+} \ln(x) = -\infty$$

We can rewrite this example as

$$\lim_{x \to 0+} \frac{\ln(x)}{\frac{1}{x}}$$

which is type $\frac{\infty}{\infty}$. L'Hôpital's Rule gives us that

$$\begin{aligned} \lim_{x \to 0^+} x \ln(x) &= \lim_{x \to 0^+} \frac{\ln(x)}{\frac{1}{x}} \\ &= \lim_{x \to 0^+} \frac{\frac{1}{x}}{\left(\frac{-1}{x^2}\right)} \\ &= 0 \end{aligned}$$

L'Hôpital's Rule

Example: Evaluate

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$$

Solution: This is type 1^∞. We write

$$\left(1+\frac{1}{x}\right)^x = e^{\ln\left(\left(1+\frac{1}{x}\right)^x\right)} = e^{x\ln\left(1+\frac{1}{x}\right)}$$

Since $x\ln\left(1+\frac{1}{x}\right)$ is type $0\cdot\infty$, we rewrite this as

$$\frac{\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}}$$

which is type $\frac{0}{0}$.

L'Hôpital's Rule

L'Hôpital's Rule gives us that

$$\begin{aligned}\lim_{x\to\infty}\frac{\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}} &= \lim_{x\to\infty}\frac{\frac{\left(\frac{-1}{x^2}\right)}{1+\frac{1}{x}}}{\left(\frac{-1}{x^2}\right)}\\ &= \frac{1}{1+\frac{1}{x}}\\ &= 1\end{aligned}$$

So

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e^{\lim_{x\to\infty}\left(\ln\left(1+\frac{1}{x}\right)^x\right)} = e$$

L'Hôpital's Rule

Problem: Show that

$$\lim_{x\to 0} \frac{4\left(e^{x^3} - 1 - x^3 - \frac{x^6}{2}\right)^2}{x^6 \tan\left(x^7\right) \sin\left(2x^5\right)} = \frac{1}{18}$$